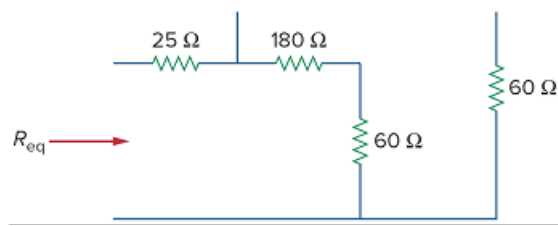


Problem

Find R_{eq} for the circuit in Fig. 2.94.

Figure 2.94:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

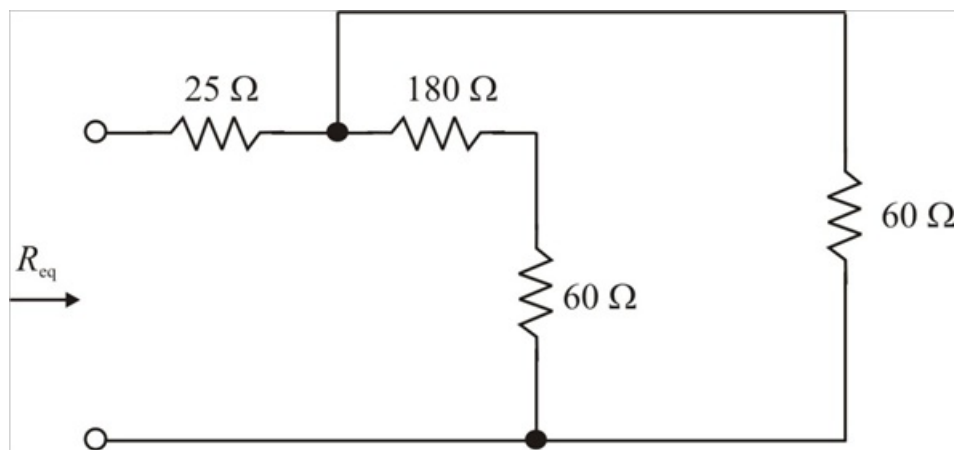


Figure 1

Step 2 of 6

To get R_{eq} , we combine the resistors in series and in parallel.

The 180 Ω and 60 Ω resistors are in series. Their combined resistance is,

$$180\Omega + 60\Omega = 240\Omega$$

Step 3 of 6

Then the reduced circuit is shown in Figure 2.

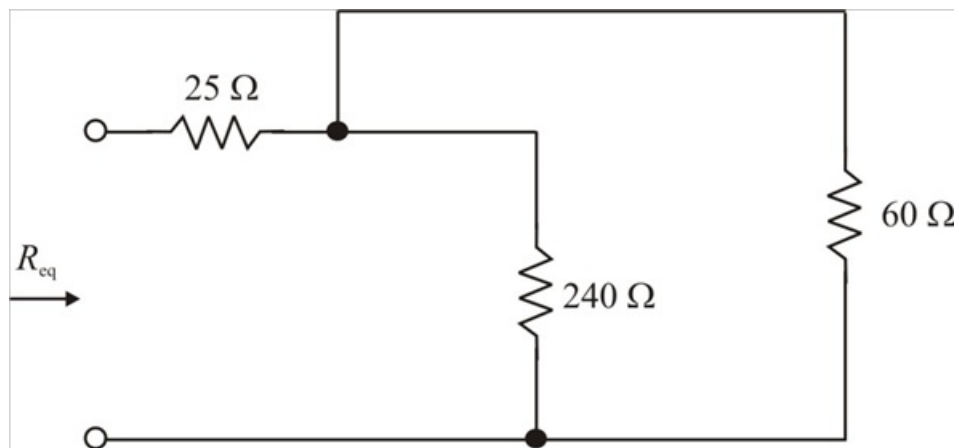


Figure 2

Step 4 of 6

The 240 Ω and 60 Ω resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 240\Omega \parallel 60\Omega &= \frac{240 \times 60}{240 + 60} \\ &= 48\Omega \end{aligned}$$

Step 5 of 6

Then the reduced circuit is shown in Figure 3.

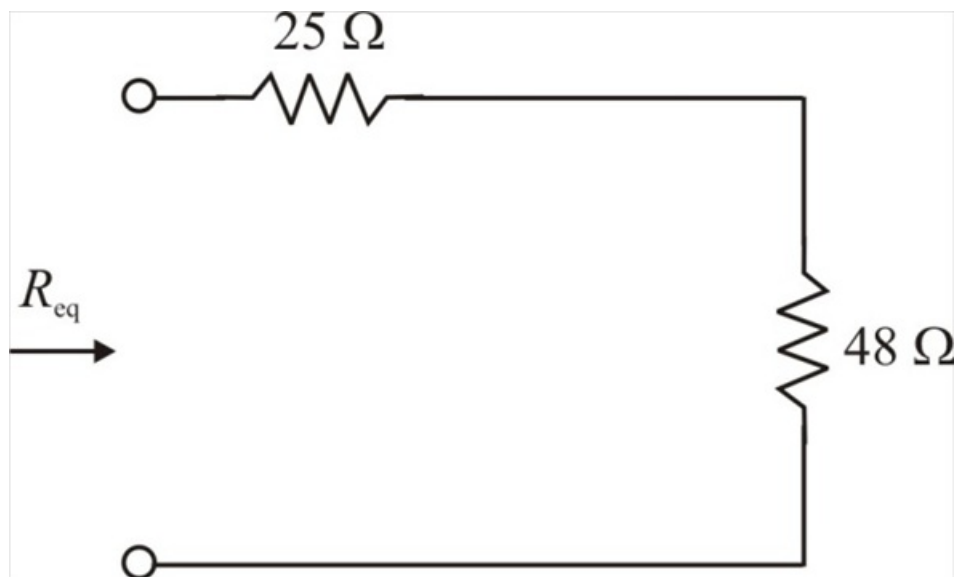


Figure 3

Step 6 of 6

The 25 Ω and 48 Ω resistors are in series. Their combined resistance is,

$$\begin{aligned} R_{eq} &= 25\Omega + 48\Omega \\ &= 73\Omega \end{aligned}$$

Therefore, the resistance R_{eq} is 73Ω.